

ORF 524 – Final Exam

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Contents

1	Question 1: Probability Theory and Asymptotics (30 points)	2
2	Question 2: Mean Absolute Error Optimality (20 points)	5
3	Question 3: Statistical Inference (50 points)	7

Notes:

- Duration of examination: *4 hours*.
Additional 20 minutes are given to upload your solutions to Canvas.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive (partial) credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Canvas.
- Only two pages of personal notes are allowed during the examination.
Basic auxiliary information about standard distributions is provided at the end of the exam.
This is otherwise a closed-book and individual examination.
Do not consult textbooks, online materials, or other sources.
Do not consult with classmates, other people or in online forums.

1 Question 1: Probability Theory and Asymptotics (30 points)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a background probability space. Recall that $\rightarrow_{\mathbb{P}}$ denotes convergence in probability and $\rightsquigarrow$ denotes convergence in distribution, with all limits taken as $n \rightarrow \infty$. In addition, recall that $\perp$ denotes (statistical) independence.

1. Let X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.

Show that $X \perp X$ if and only if X is (a.s.) constant (i.e., $\mathbb{P}[X = c] = 1$, for some $c \in \mathbb{R}$.)

- **Solution:** First, assume $\mathbb{P}[X = c] = 1$. Then, for any $A, B \in \mathcal{B}$,

$$\mathbb{P}[X \in A, X \in B] = \mathbb{1}(c \in A, c \in B) = \mathbb{1}(c \in A)\mathbb{1}(c \in B) = \mathbb{P}[X \in A]\mathbb{P}[X \in B],$$

which gives the “if” direction. Second, assume $X \perp X$. Then,

$$\mathbb{P}(X \leq x) = \mathbb{P}(X \leq x, X \leq x) = \mathbb{P}(X \leq x)\mathbb{P}(X \leq x) = [\mathbb{P}(X \leq x)]^2,$$

which implies $F(x) = \mathbb{P}(X \leq x) \in \{0, 1\}$. Thus, since $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow +\infty} F(x) = 1$, there must exist $c \in \mathbb{R}$ such that $\mathbb{P}(X < c) = 0$ and $\mathbb{P}(X \leq c) = 1$, which gives the “only if” direction.

2. Let (X, Y, Z) be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$.

Show that $\mathbb{E}[X\mathbb{E}[Y|Z]] = \mathbb{E}[Y\mathbb{E}[X|Z]]$, whenever both integrals exist.

- **Solution:** Using the law of iterated expectations, and assuming all integrals exist,

$$\mathbb{E}[X\mathbb{E}[Y|Z]] = \mathbb{E}[\mathbb{E}[X\mathbb{E}[Y|Z]|Z]] = \mathbb{E}[\mathbb{E}[X|Z]\mathbb{E}[Y|Z]] = \mathbb{E}[\mathbb{E}[Y\mathbb{E}[X|Z]|Z]] = \mathbb{E}[Y\mathbb{E}[X|Z]],$$

and the result follows.

3. Let $(X_n : n \geq 1)$ be a sequence of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$.

Show that $|X_n|\mathbb{1}(|X_n| \geq M) \rightarrow_{\mathbb{P}} 0$ for all $M > 0$ whenever $\mathbb{E}[|X_n|^{1+\delta}] \rightarrow 0$ for some $\delta > 0$.

- **Solution:** Note that

$$\begin{aligned} \mathbb{E}[|X_n|\mathbb{1}(|X_n| \geq M)] &= \mathbb{E}[|X_n|^{1+\delta}|X_n|^{-\delta}\mathbb{1}(|X_n| \geq M)] \\ &\leq M^{-\delta}\mathbb{E}[|X_n|^{1+\delta}\mathbb{1}(|X_n| \geq M)] \\ &\leq M^{-\delta}\mathbb{E}[|X_n|^{1+\delta}] \rightarrow 0, \end{aligned}$$

for any $M > 0$. The result now follows from Markov inequality.

4. Let $(X_i : 1 \leq i \leq n)$ be a random sample with $X_i \sim \mathbb{P}$ and $\mathbb{E}[|X_i|^4] < \infty$. Let $\mu = \mathbb{E}[X_i]$, $\sigma^2 = \mathbb{V}[X_i] = \mathbb{E}[(X_i - \mu)^2]$, and $\mathbb{C}[X_i, X_i^2] = \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_i^2 - \mathbb{E}[X_i^2])]$.

(a) Show that

$$\sqrt{n} \left(\begin{bmatrix} \frac{1}{n} \sum_{i=1}^n X_i \\ \frac{1}{n} \sum_{i=1}^n X_i^2 \end{bmatrix} - \begin{bmatrix} \mu \\ \sigma^2 + \mu^2 \end{bmatrix} \right) \rightsquigarrow \mathbf{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \mathbb{V}[X_i] & \mathbb{C}[X_i, X_i^2] \\ \mathbb{C}[X_i, X_i^2] & \mathbb{V}[X_i^2] \end{bmatrix} \right).$$

- **Solution:** Let $\mathbf{W}_i = (X_i, X_i^2)' \in \mathbb{R}^2$. Then, $\mathbb{E}[\mathbf{W}_i] = \boldsymbol{\mu}_W = (\mathbb{E}[X_i], \mathbb{E}[X_i^2])' = (\mu, \sigma^2 + \mu^2)'$ and

$$\mathbb{V}[\mathbf{W}_i] = \boldsymbol{\Sigma} = \begin{bmatrix} \mathbb{V}[X_i] & \mathbb{C}[X_i, X_i^2] \\ \mathbb{C}[X_i, X_i^2] & \mathbb{V}[X_i^2] \end{bmatrix} \in \mathbb{R}^{2 \times 2},$$

for all $i = 1, \dots, n$. Therefore, by i.i.d and finite fourth moment of X_i , we can apply the CLT to obtain: $\sqrt{n}(\bar{\mathbf{W}}_n - \boldsymbol{\mu}_W) \rightsquigarrow \mathbf{N}(\mathbf{0}, \boldsymbol{\Sigma})$, where $\bar{\mathbf{W}}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{W}_i = (\frac{1}{n} \sum_{i=1}^n X_i, \frac{1}{n} \sum_{i=1}^n X_i^2)'$.

(b) Show that

$$\sqrt{n} \left(\begin{bmatrix} \bar{X}_n \\ S_n^2 \end{bmatrix} - \begin{bmatrix} \mu \\ \sigma^2 \end{bmatrix} \right) \rightsquigarrow \mathbf{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma^2 & \mathbb{E}[(X_i - \mu)^3] \\ \mathbb{E}[(X_i - \mu)^3] & \mathbb{E}[(X_i - \mu)^4] - \sigma^4 \end{bmatrix} \right).$$

- **Solution:** Define the function $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ as $g(x, y) = (x, y - x^2)'$, which is twice continuous, with gradient:

$$\nabla g(x, y) = \begin{bmatrix} 1 & 0 \\ -2x & 1 \end{bmatrix}.$$

Noting that $\nabla g(\boldsymbol{\mu}_W) \neq \mathbf{0}$, we can apply the delta method, which gives

$$\sqrt{n}(g(\bar{\mathbf{W}}_n) - g(\boldsymbol{\mu}_W)) \rightsquigarrow \mathbf{N}(\mathbf{0}, \nabla g(\boldsymbol{\mu}_W) \boldsymbol{\Sigma} \nabla g(\boldsymbol{\mu}_W)'),$$

where

$$\nabla g(\boldsymbol{\mu}_W) \boldsymbol{\Sigma} \nabla g(\boldsymbol{\mu}_W)' = \begin{bmatrix} 1 & 0 \\ -2\mu & 1 \end{bmatrix} \begin{bmatrix} \mathbb{V}[X_i] & \mathbb{C}[X_i, X_i^2] \\ \mathbb{C}[X_i, X_i^2] & \mathbb{V}[X_i^2] \end{bmatrix} \begin{bmatrix} 1 & -2\mu \\ 0 & 1 \end{bmatrix}.$$

Finally, using $\mathbb{V}[X_i^2] = \mathbb{E}[X_i^4] - (\sigma^2 + \mu^2)^2$ and multiplying out the three matrices

$$\nabla g(\boldsymbol{\mu}_W) \boldsymbol{\Sigma} \nabla g(\boldsymbol{\mu}_W)' = \begin{bmatrix} \sigma^2 & \mathbb{C}[X_i, X_i^2] - 2\mu\sigma^2 \\ \mathbb{C}[X_i, X_i^2] - 2\mu\sigma^2 & \mathbb{E}[X_i^4] - \sigma^4 - 4\mu\mathbb{C}[X_i, X_i^2] + 2\mu^2\sigma^2 - \mu^4 \end{bmatrix},$$

and the result follows because $\mathbb{E}[(X_i - \mu)^3] = \mathbb{C}[X_i, X_i^2] - 2\mu\sigma^2$ and $\mathbb{E}[(X_i - \mu)^4] = \mathbb{E}[X_i^4] - 4\mu\mathbb{C}[X_i, X_i^2] + 2\mu^2\sigma^2 - \mu^4$.

- (c) Give necessary and sufficient conditions for $\bar{X}_n$ and S_n^2 to be asymptotically independent. Give an example where $\bar{X}_n$ and S_n^2 are independent for all $n \geq 1$.

- **Solution:** A necessary and sufficient assumption for asymptotic independence is $\mathbb{E}[(X_i - \mu)^3] = 0$, which is verified if X_i 's are symmetric (around their mean) random variables. Thus, asymptotic independence does not hold in general. Of course, they are independent for all $n \geq 1$ (and hence asymptotically) if $X_i \sim \mathcal{N}(\mu, \sigma^2)$.

2 Question 2: Mean Absolute Error Optimality (20 points)

This question studies best Mean Absolute Error (MAE) prediction. Let $(Y, X) \in \mathbb{R}^2$ be an integrable random vector with joint probability law $\mathbb{P}$. The integrability assumption is imposed only for simplicity, but it is not really needed, and notice that no interchanging of differentiability and integration will be used. Let $\mathbb{P}_Y$ be the marginal probability law of Y , and $\mathbb{P}_{Y|X}$ be the (regular) conditional probability law of $Y|X$.

1. Show that if $\mathbb{P}_Y[Y \leq a] \geq 1/2$ and $a \leq b$, then for all $a, b \in \mathbb{R}$,

$$\mathbb{E}[|Y - b|] - \mathbb{E}[|Y - a|] \geq (a - b) \cdot (1 - 2 \cdot \mathbb{P}_Y[Y \leq a]) \geq 0.$$

Hint: $|Y - b| - |Y - a| = 2 \cdot (b - Y) \cdot \mathbb{1}(a < Y < b) + (a - b) \cdot [\mathbb{1}(Y > a) - \mathbb{1}(Y \leq a)]$.

- **Solution:** First, note that for any $y \in \mathbb{R}$,

$$|Y - y| = (Y - y)\mathbb{1}(Y > y) + (y - Y)\mathbb{1}(Y \leq y) = (Y - y)\mathbb{1}(Y \geq y) + (y - Y)\mathbb{1}(Y < y).$$

This gives, for $a < b$,

$$\begin{aligned} |Y - b| - |Y - a| &= (Y - b)\mathbb{1}(Y \geq b) + (b - Y)\mathbb{1}(Y < b) - (Y - a)\mathbb{1}(Y > a) - (a - Y)\mathbb{1}(Y \leq a) \\ &= Y[\mathbb{1}(Y \geq b) - \mathbb{1}(Y > a) + \mathbb{1}(Y \leq a) - \mathbb{1}(Y < b)] \\ &\quad + b[\mathbb{1}(Y < b) - \mathbb{1}(Y \geq b)] + a[\mathbb{1}(Y > a) - \mathbb{1}(Y \leq a)] \\ &= -2Y\mathbb{1}(a < Y < b) + b[\mathbb{1}(Y < b) - \mathbb{1}(Y \leq a) - \mathbb{1}(Y \geq b) + \mathbb{1}(Y > a)] \\ &\quad + (a - b)[\mathbb{1}(Y > a) - \mathbb{1}(Y \leq a)] \\ &= 2(b - Y)\mathbb{1}(a < Y < b) + (a - b)[\mathbb{1}(Y > a) - \mathbb{1}(Y \leq a)], \end{aligned}$$

where the second equality is simply rearranging the first line, the third equality follows from $\mathbb{1}(Y \geq b) - \mathbb{1}(Y > a) = \mathbb{1}(Y \leq a) - \mathbb{1}(Y < b) = -\mathbb{1}(a < Y < b)$ and by adding and subtracting $b[\mathbb{1}(Y > a) - \mathbb{1}(Y \leq a)]$, and the fourth equality is rearranging. Moreover, when $a = b$ the result also trivially holds. This establishes the hint. Given the equality in the hint, the fact that $2(b - Y)\mathbb{1}(a < Y < b) \geq 0$, and taking expectations both sides, we obtain

$$\begin{aligned} \mathbb{E}[|Y - b|] - \mathbb{E}[|Y - a|] &\geq (a - b)[\mathbb{P}_Y(Y > a) - \mathbb{P}_Y(Y \leq a)] \\ &= (a - b)(1 - 2\mathbb{P}_Y[Y \leq a]) \geq 0, \end{aligned}$$

using the assumption $\mathbb{P}_Y(Y \leq a) \geq 1/2$ and $a \leq b$.

2. Let m be a median of $\mathbb{P}_Y$, that is, $\mathbb{P}_Y[Y \leq m] \geq 1/2$ and $\mathbb{P}_Y[Y \geq m] \geq 1/2$.

Show that

$$\mathbb{E}[|Y - a|] \leq \mathbb{E}[|Y - b|].$$

for all $a, b \in \mathbb{R}$ such that $m \leq a \leq b$ or $m \geq a \geq b$,

- **Solution:** consider first the case $m \leq a \leq b$. We have that $\mathbb{P}_Y(Y \leq a) \geq \mathbb{P}_Y(Y \leq m) \geq 1/2$ and $a \leq b$, so by Part 1, $\mathbb{E}[|Y - b|] \geq \mathbb{E}[|Y - a|]$. Next, consider the second case, $m \geq a \geq b$. We have that $\mathbb{P}_Y(Y \geq a) \geq \mathbb{P}_Y(Y \geq m) \geq 1/2$ and therefore $\mathbb{P}(-Y \leq -a) \geq 1/2$. Since $-b \geq -a$, we can apply Part 1 to $-a$, $-b$ and $-Y$ to get:

$$\mathbb{E}[|(-Y) - (-a)|] \leq \mathbb{E}[|(-Y) - (-b)|] \iff \mathbb{E}[|Y - a|] \leq \mathbb{E}[|Y - b|].$$

3. Let $m(X)$ be a median of $\mathbb{P}_{Y|X}$. Show that

$$m(X) \in \arg \min_{T \in \mathcal{T}} \mathbb{E}[|Y - T(X)|],$$

where $\mathcal{T} = \{T(\cdot) : \mathbb{E}[|T(X)|] < \infty\}$.

- **Solution:** Part 2 implies that if m is a median of $\mathbb{P}_Y$, then for any $b \in \mathbb{R}$, $\mathbb{E}[|Y - m|] \leq \mathbb{E}[|Y - b|]$. Because these results hold for conditional probabilities too, we get that if $m(X)$ is a conditional median, then:

$$\mathbb{E}[|Y - m(X)| \mid X] \leq \mathbb{E}[|Y - T(X)| \mid X] \quad (\text{a.s.}) \quad \forall T(\cdot) \in \mathcal{T}.$$

Integrating over the distribution of X on both sides, it follows by the law of iterated expectations that $\mathbb{E}[|Y - m(X)|] \leq \mathbb{E}[|Y - T(X)|]$ for all $T(\cdot) \in \mathcal{T}$.

4. Suppose $(Y, X)' \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_Y \\ \mu_X \end{bmatrix} \quad \text{and} \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{YY} & \sigma_{YX} \\ \sigma_{XY} & \sigma_{XX} \end{bmatrix}$$

Find m and $m(X)$. Are they (a.s.) unique?

- **Solution:** because the Normal distribution is symmetric, its median is equal to its mean. Thus, we have that

$$m = \mu_Y \quad \text{and} \quad m(X) = \mathbb{E}[Y|X] = \mu_Y + \frac{\sigma_{XY}}{\sigma_{XX}}(X - \mu_X),$$

which are a.s. unique. In this case, $m = \mu_Y = \mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[m(X)]$, but in general it is not true that $m = \mathbb{E}[m(X)]$.

3 Question 3: Statistical Inference (50 points)

Suppose $X_1, X_2, \dots, X_n$ is an i.i.d. sequence from $X \sim \mathbb{P}_\theta$, where $X_i \in \{0, 1\}$ for all i and $\theta = p_0 = \mathbb{P}_{p_0}[X_i = 1] \in (0, 1)$. Recall that $\rightarrow_{\mathbb{P}}$ denotes convergence in probability and $\rightsquigarrow$ denotes convergence in distribution. All limits are taken as $n \rightarrow \infty$, unless explicitly noted otherwise.

1. Show that the (proportion) estimator

$$\hat{p}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

is both a method of moments and the maximum likelihood estimator of p_0 .

- **Solution:** Since $\mathbb{P}_{p_0}[X_i = 1] = \mathbb{E}_{p_0}[X_i]$, it follows by the plug-in principle that $\hat{p}_n = \frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{n} \sum_{i=1}^n \mathbb{1}(X_i = 1)$ is a MME. Also, because $X_i \sim \text{Bernoulli}(p_0)$, it follows that $\hat{p}_n$ is the MLE of p_0 .

2. Show that $\hat{p}_n$ is unbiased, and that $\mathbb{V}[\hat{p}_n] = p_0(1 - p_0)/n$. Is $\hat{p}_n$ a UMVUE of p_0 ?

- **Solution:** We have $\mathbb{E}_{p_0}[\hat{p}_n] = \mathbb{E}_{p_0}[X_i] = \mathbb{P}_{p_0}[X_i = 1] = p_0$ and $\mathbb{V}_{p_0}[\hat{p}_n] = \mathbb{V}_{p_0}[X_i]/n = p_0(1 - p_0)/n$. As discussed in class multiple times, $\hat{p}_n$ is the (a.s. unique) UMVUE of p_0 . This follows because $\hat{p}_n$ is based on a complete sufficient statistic and is unbiased for p_0 . As also shown in class, $\mathbb{V}_{p_0}[\hat{p}_n] = \text{CRLB}(p_0)$.

3. Providing sufficient conditions, propose an estimator $\hat{V}_n$ satisfying $\hat{V}_n = \mathbb{V}[\hat{p}_n] + O_{\mathbb{P}}(n^{-3/2})$.

- **Solution:** Consider the plug-in estimator $\hat{V}_n = \hat{p}_n(1 - \hat{p}_n)/n$. Then,

$$\begin{aligned} \mathbb{V}[\hat{p}_n] - \hat{V}_n &= \frac{p_0(1 - p_0)}{n} - \frac{\hat{p}_n(1 - \hat{p}_n)}{n} = \frac{p_0 - \hat{p}_n}{n} - \frac{p_0^2 - \hat{p}_n^2}{n} \\ &= \frac{p_0 - \hat{p}_n}{n} - \frac{(p_0 - \hat{p}_n)(p_0 + \hat{p}_n)}{n} = O_{\mathbb{P}}(n^{-3/2}), \end{aligned}$$

because $\hat{p}_n - p_0 = O_{\mathbb{P}}(n^{-1/2})$.

4. Providing sufficient conditions, show that

$$T_n := \frac{\hat{p}_n - p_0}{\sqrt{\hat{V}_n}} \rightsquigarrow Z,$$

where $Z \sim \mathcal{N}(0, 1)$.

- **Solution:**

$$\frac{\hat{p}_n - p_0}{\sqrt{\hat{V}_n}} = \frac{\sqrt{n}(\hat{p}_n - p_0)}{\sqrt{\hat{p}_n(1 - \hat{p}_n)}} = \sqrt{\frac{p_0(1 - p_0)}{\hat{p}_n(1 - \hat{p}_n)}} \cdot \frac{\sqrt{n}(\bar{X}_n - p_0)}{\sqrt{p_0(1 - p_0)}}$$

Now, by the CLT, given finite second moments,

$$\frac{\sqrt{n}(\bar{X}_n - p_0)}{\sqrt{p_0(1-p_0)}} \rightsquigarrow \mathbf{N}(0, 1),$$

and by the CMT,

$$\sqrt{\frac{p_0(1-p_0)}{\hat{p}_n(1-\hat{p}_n)}} \rightarrow_{\mathbb{P}} 1$$

so the required result follows by Slutsky's theorem.

5. Show that $\mathbb{P}_{p_0}[p_0 \in \text{CI}_{1-\alpha}] \rightarrow 1 - \alpha$, where

$$\text{CI}_{1-\alpha} = \left[\hat{p}_n - \Phi^{-1}(1 - \alpha/2) \cdot \sqrt{\frac{\hat{p}_n(1 - \hat{p}_n)}{n}}, \hat{p}_n + \Phi^{-1}(1 - \alpha/2) \cdot \sqrt{\frac{\hat{p}_n(1 - \hat{p}_n)}{n}} \right]$$

with $\Phi(t) := \mathbb{P}[Z \leq t] = \int_{-\infty}^t (2\pi)^{-1/2} e^{-u^2/2} du$.

- **Solution:** The large sample Gaussian approximation justifies the usual $100(1 - \alpha)\%$ confidence interval for the unknown proportion p_0 because

$$\begin{aligned} \mathbb{P}_{p_0}[p_0 \in \text{CI}_{1-\alpha}] &= \mathbb{P}_{p_0} \left[-\Phi^{-1}(1 - \alpha/2) \leq \frac{\hat{p}_n - p_0}{\sqrt{\hat{V}_n}} \leq \Phi^{-1}(1 - \alpha/2) \right] \\ &\rightarrow \mathbb{P}_{p_0} [-\Phi^{-1}(1 - \alpha/2) \leq Z \leq \Phi^{-1}(1 - \alpha/2)] = 1 - \alpha. \end{aligned}$$

6. A quantity often used in categorical data analysis is the “log-odds”:

$$\zeta_0 = \log \left(\frac{p_0}{1 - p_0} \right).$$

Construct an asymptotically valid $100(1 - \alpha)\%$ confidence interval for ζ_0 , where $\alpha \in (0, 1)$.

- **Solution:** The transformation $g(x) = \log(x/(1 - x))$ is continuous (for $x \in (0, 1)$), and $g'(x) = \frac{1}{p(1-p)}$. So by the previous result and the delta method:

$$\sqrt{n}(\hat{\zeta}_n - \zeta_0) \rightarrow_d \mathbf{N}(0, g'(p_0)^2 p_0(1 - p_0)) =_d \mathbf{N} \left(0, \frac{1}{p_0(1 - p_0)} \right)$$

where $\hat{\zeta}_n = \log(\hat{p}_n/(1 - \hat{p}_n))$. Hence, the confidence interval is given by:

$$\left[\log \left(\frac{\hat{p}_n}{1 - \hat{p}_n} \right) \pm \Phi^{-1}(1 - \alpha/2) \sqrt{\frac{1}{n\hat{p}_n(1 - \hat{p}_n)}} \right].$$

7. The confidence intervals above are only asymptotically valid *pointwise* in $\theta = p_0 \in (0, 1)$.

However, their confidence coefficients are zero! That is, for $\text{CI}_{1-\alpha}$ above, we have

$$\inf_{p_0 \in (0,1)} \mathbb{P}_{p_0} [p_0 \in \text{CI}_{1-\alpha}] = 0.$$

Show this result using the following two steps.

(a) Show that, for each fixed n ,

$$\mathbb{P}_{p_0} [p_0 \in \text{CI}_{1-\alpha}] = 1 - \mathbb{P}_{p_0} [p_0 \notin \text{CI}_{1-\alpha}] \leq 1 - \mathbb{P}_{p_0} [X_1 = 0, X_2 = 0, \dots, X_n = 0].$$

• **Solution:** We are asked to show that

$$1 - \mathbb{P}_{p_0} [p_0 \notin \text{CI}_{1-\alpha}] \leq 1 - \mathbb{P}_{p_0} [X_1 = 0, X_2 = 0, \dots, X_n = 0],$$

which is equivalent to showing

$$\mathbb{P}_{p_0} [p_0 \notin \text{CI}_{1-\alpha}] \geq \mathbb{P}_{p_0} [X_1 = 0, X_2 = 0, \dots, X_n = 0]$$

In words, the idea is that when all the X_i 's are zero, the confidence interval collapses to the singleton zero, and since $p_0 > 0$, the probability of the interval containing the true parameter will be zero. More formally,

$$X_i = 0, \forall i \quad \Rightarrow \quad \hat{p}_n = 0 \quad \Rightarrow \quad \text{CI}_{1-\alpha} = \{0\} \quad \Rightarrow \quad p_0 \notin \text{CI}_{1-\alpha}$$

which means that

$$\{\omega \in \Omega : X_i = 0, \forall i\} \subseteq \{\omega \in \Omega : p_0 \notin \text{CI}_{1-\alpha}\}$$

and hence:

$$\mathbb{P}_{p_0} [X_1 = 0, X_2 = 0, \dots, X_n = 0] \leq \mathbb{P}_{p_0} [p_0 \notin \text{CI}_{1-\alpha}]$$

by monotonicity of probability measures.

(b) Conclude that the confidence coefficient of $\text{CI}_{1-\alpha}$ is zero in finite samples, and hence asymptotically.

• **Solution:** because, under $\mathbb{P}_{p_0}$, $X_i \stackrel{iid}{\sim} \text{Bernoulli}(p_0)$, we have that

$$\mathbb{P}_{p_0} [X_1 = 0, X_2 = 0, \dots, X_n = 0] = (1 - p_0)^n$$

Thus, from (a),

$$0 \leq \mathbb{P}_{p_0} [p_0 \in \text{CI}_{1-\alpha}] \leq 1 - \mathbb{P}_{p_0} [X_1 = 0, X_2 = 0, \dots, X_n = 0] = 1 - (1 - p_0)^n$$

for all $p_0 \in (0, 1)$, which gives

$$0 \leq \inf_{p_0 \in (0,1)} \mathbb{P}_{p_0}[p_0 \in \text{CI}_{1-\alpha}] \leq \inf_{p_0 \in (0,1)} \{1 - (1 - p_0)^n\},$$

and therefore

$$\inf_{p_0 \in (0,1)} \{1 - (1 - p_0)^n\} = 1 - \sup_{p_0 \in (0,1)} (1 - p_0)^n = 1 - 1 = 0,$$

giving the result.

8. To better understand the previous result, let $(B_n : n \geq 1)$ be a sequence of random variables with $B_n \sim \text{Binomial}(n, p)$, where $n \in \mathbb{N}$ and $p \in (0, 1)$.

(a) Show that

$$\frac{B_n - np}{\sqrt{np(1-p)}} \rightsquigarrow \mathbf{N}(0, 1),$$

which gives the well-known Gaussian approximation to the Binomial distribution.

This approximation is valid for “not too small” p , in the sense that $np \rightarrow \infty$.

- **Solution:** Think of B_n as a sum of n independent Bernoulli random variables X_i , that is, set $B_n = \sum_{i=1}^n X_i$ where $X_i \stackrel{iid}{\sim} \text{Bernoulli}(p)$. Then,

$$\frac{B_n - np}{\sqrt{np(1-p)}} = \sum_{i=1}^n \frac{X_i - p}{\sqrt{np(1-p)}} = \frac{\sqrt{n}(\bar{X}_n - p)}{\sqrt{p(1-p)}}$$

and the result follows by the CLT, as in part 4.

- (b) Suppose now that p is “small” in the sense that $np \rightarrow \lambda \in (0, \infty)$. Note this forces $p \rightarrow 0$ as $n \rightarrow \infty$. Use this embedding to show that

$$B_n \rightsquigarrow \text{Poisson}(\lambda),$$

which corresponds to the “law of rare events” approximation.

- **Solution:** Because both probabilities are discrete, we can show this by proving convergence of the probability mass functions. Start by noting that, since $np \rightarrow \lambda$, then $np = \lambda + o(1)$ and hence $p = \lambda/n + o(n^{-1})$. Let $k \in \mathbb{N}$, then we have:

$$\begin{aligned} \mathbb{P}[B_n = k] &= \binom{n}{k} p^k (1-p)^{n-k} \\ &= \binom{n}{k} (1-p)^n \cdot \frac{1}{n^k} \left(\frac{np}{1-p} \right)^k \\ &= \binom{n}{k} \frac{1}{n^k} \cdot \left(1 - \frac{\lambda}{n} + o(n^{-1}) \right)^n \cdot \left(\frac{\lambda}{1 - \lambda/n} + o(1) \right)^k \end{aligned}$$

Now, for the first term;

$$\begin{aligned}\binom{n}{k} \frac{1}{n^k} &= \frac{n!}{(n-k)!k!} \cdot \frac{1}{n^k} = \frac{n(n-1) \cdots (n-k+1)}{n^k} \cdot \frac{1}{k!} \\ &= \frac{n}{n} \cdot \left(\frac{n-1}{n}\right) \cdots \left(\frac{n-k+1}{n}\right) \cdot \frac{1}{k!} \rightarrow \frac{1}{k!}\end{aligned}$$

For the second term, the intuition is that for n large enough, the $o(1)$ term is negligible so that $(1 - \lambda/n + o(1))$ behaves like $(1 - \lambda/n)$ and thus the term should converge to $e^{-\lambda}$. To show this formally, take logs and use a first-order Taylor expansion around 1:

$$\begin{aligned}\log \left[\left(1 - \frac{\lambda}{n} + o(1) \right)^n \right] &= n \log \left(1 - \frac{\lambda}{n} + o(n^{-1}) \right) \\ &= n \left(-\frac{\lambda}{n} + o(n^{-1}) + o\left(\frac{\lambda}{n}\right) \right) \\ &= -\lambda + o(1) \\ &\rightarrow -\lambda\end{aligned}$$

and hence

$$\left(1 - \frac{\lambda}{n} + o(1) \right)^n \rightarrow e^{-\lambda}$$

Finally, for the third term, it is straightforward to see that

$$\left(\frac{\lambda}{1 - \lambda/n} + o(1) \right)^k \rightarrow \lambda^k$$

Putting all the results together,

$$\mathbb{P}[B_n = k] \rightarrow \frac{1}{k!} \cdot e^{-\lambda} \cdot \lambda^k = \frac{\lambda^k}{k!} e^{-\lambda}$$

which is the p.d.f. of a Poisson random variable.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, $(n > 1)$
	Variance	$n/(n-2)$, $(n > 2)$
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, $(m > 2)$
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, $(m > 4)$
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, $(\theta > 1)$
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, $(\theta > 2)$
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$